HUMAN-ENVIRONMENT INTERACTION

PROJECT PROPOSAL

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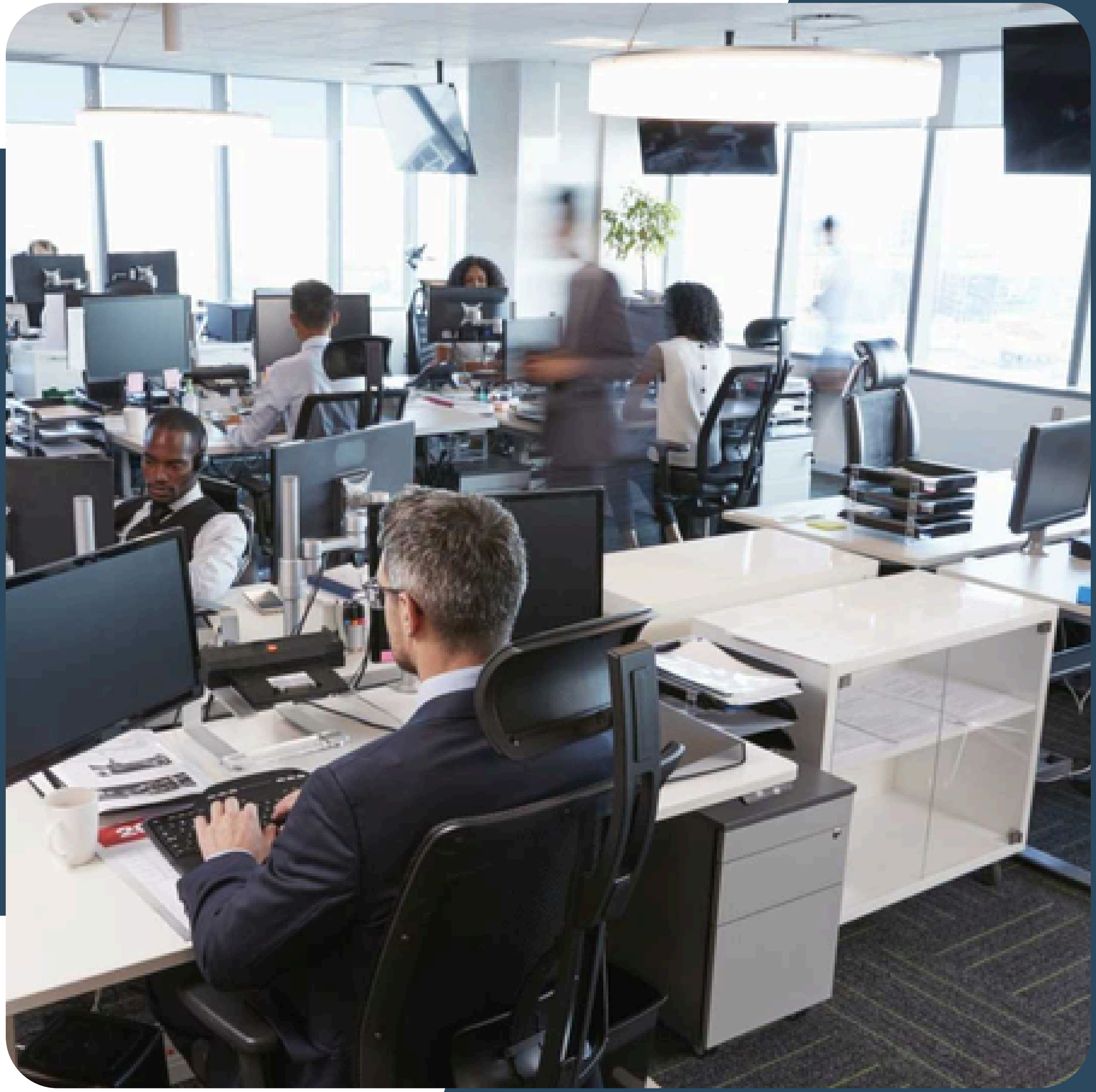
PROBLEM ANALYSIS

A new office building has been designed with high-tech systems that control lighting, temperature, and air quality. However, employees feel stressed and uncomfortable because the systems are too rigid, offering little control over personal preferences and failing to respond dynamically to environmental changes.



POTENTIAL PROBLEMS

- Offering little control
- Not Adapting to Changes in the Environment
- Overcomplication-Related Stress
- Different Preferences
- The absence of Feedback Systems

Solutions

Offering little control

To ensure personal control and lower stress, employees can use a straightforward interface like app or touch panels to customize or control the lighting, temperature, and air quality.

Solutions

Not Adapting to Changes in the Environment

The system uses real-time sensors to monitor environmental factors like temperature, occupancy, and sunlight, ensuring responsiveness and automatic adjustment of shared spaces. For example what is the time, date and the weather of a specific environment.

Solutions

Overcomplication-Related Stress

Icons like chat box, voice command, dark mode, keyboard, on and off button can make their life more easier. Like pre-set settings like "Focus" or "Relax" that can lessen the stress for staff members. In other words make the system user friendly.

Solutions

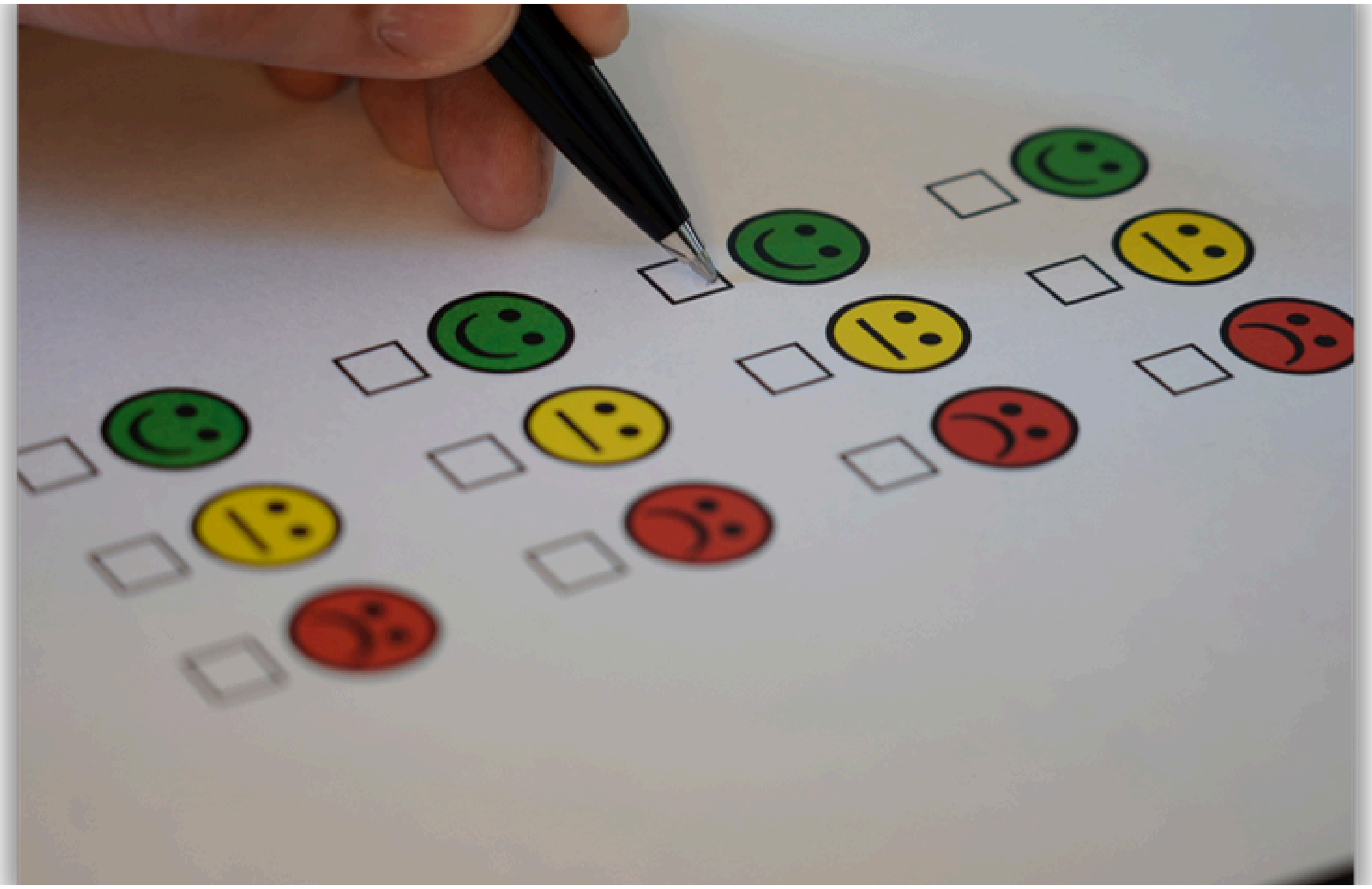
Different Preferences

Most employees have different preferences allowing them all to gain access is one of the solution. With the help of AI, we can learn and predict user habits, adjusts settings dynamically based on those patterns, and thereby can enhance adaptability.

Solutions

The absence of Feedback Systems

Employee input is gathered using an integrated feedback function, which helps the system to evolve and adjust to user requirements. Solve real time errors and ensure overall good quality performance.



HCI PRINCIPLES

01. USABILITY

The system ensures effectiveness, efficiency like how easy to use and ensures safety of the employees.

02. ACCESSIBILITY

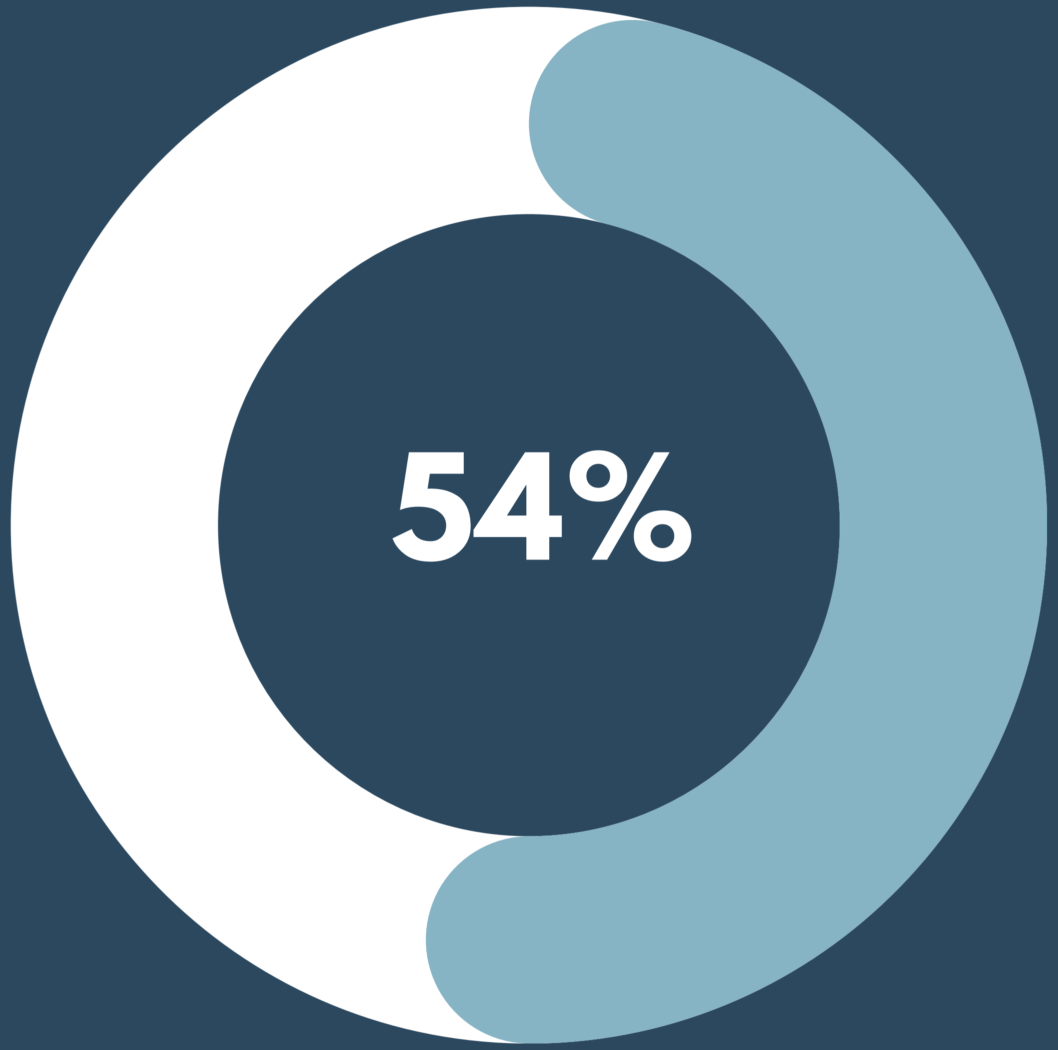
The system is accessible by everyone in the workplace. All employees can utilize the system with or without disabilities.

03. ETHICAL CONSIDERATIONS

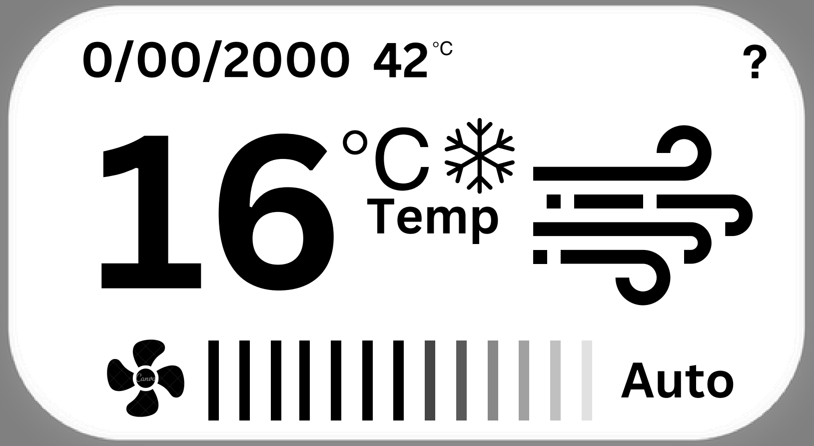
The system considers well-being and comfort for the employees with rights, respect,fairness and security or privacy at everyone in the workplace which can foster a supportive work environment.

STATISTICS

Internet of Things (IoT)is a network of physical objects that are connected and exchange information over the internet with no human interaction. They are these devices that should possess sensors, software, and other technologies that can facilitate collection, processing and sharing of information.



According to Duggal (2024). Kevin Ashton created the phrase "Internet of Things" (IoT) in 1999. But it was not until Gartner added IoT to its list of new emerging technologies in 2011 that it began to garner worldwide traction. As of **2021**, the globe had **21.7 billion active connected devices**, with IoT devices accounting for more than **11.7 billion (54 percent)**. This signifies that the globe has more IoT devices than non-IoT devices .



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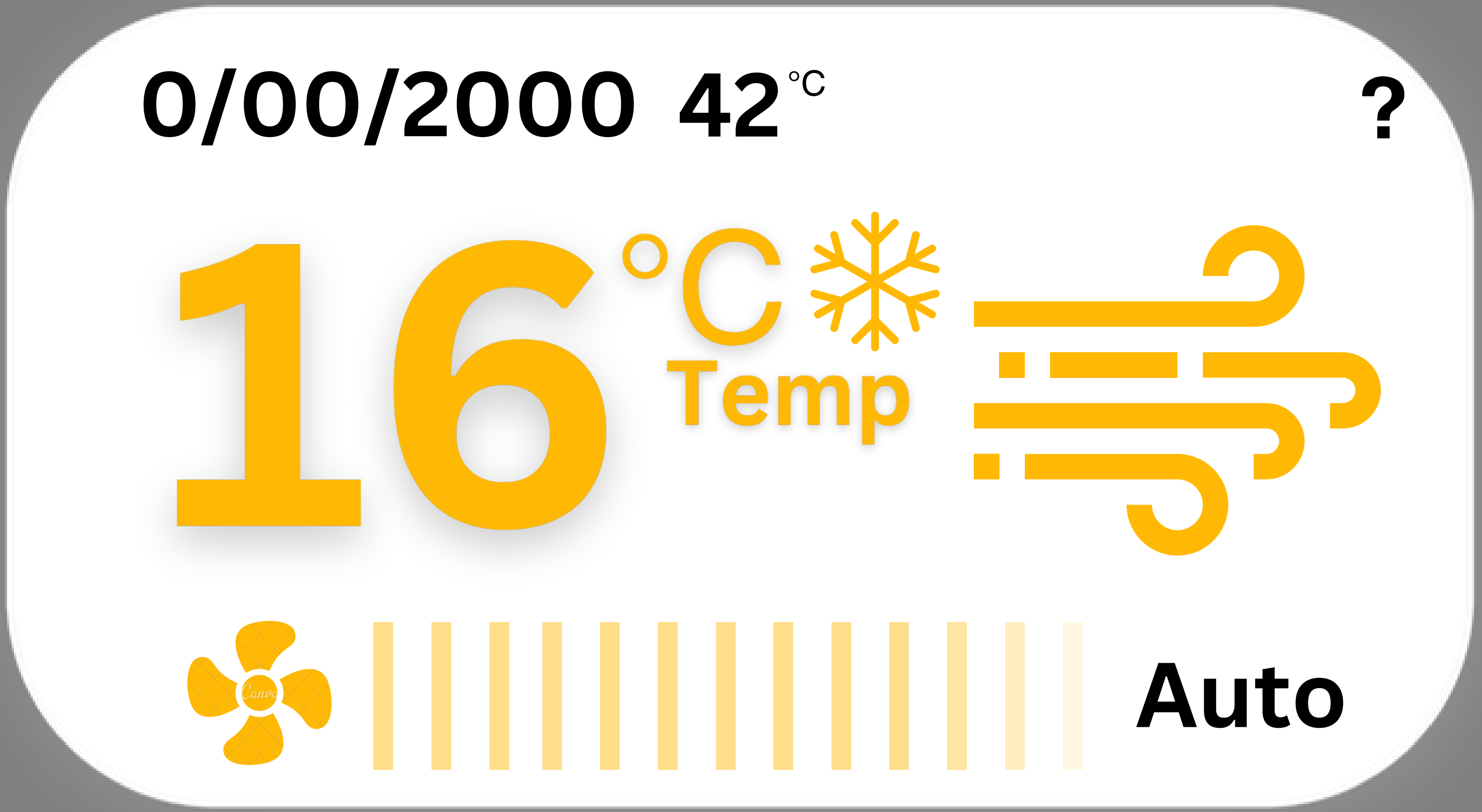
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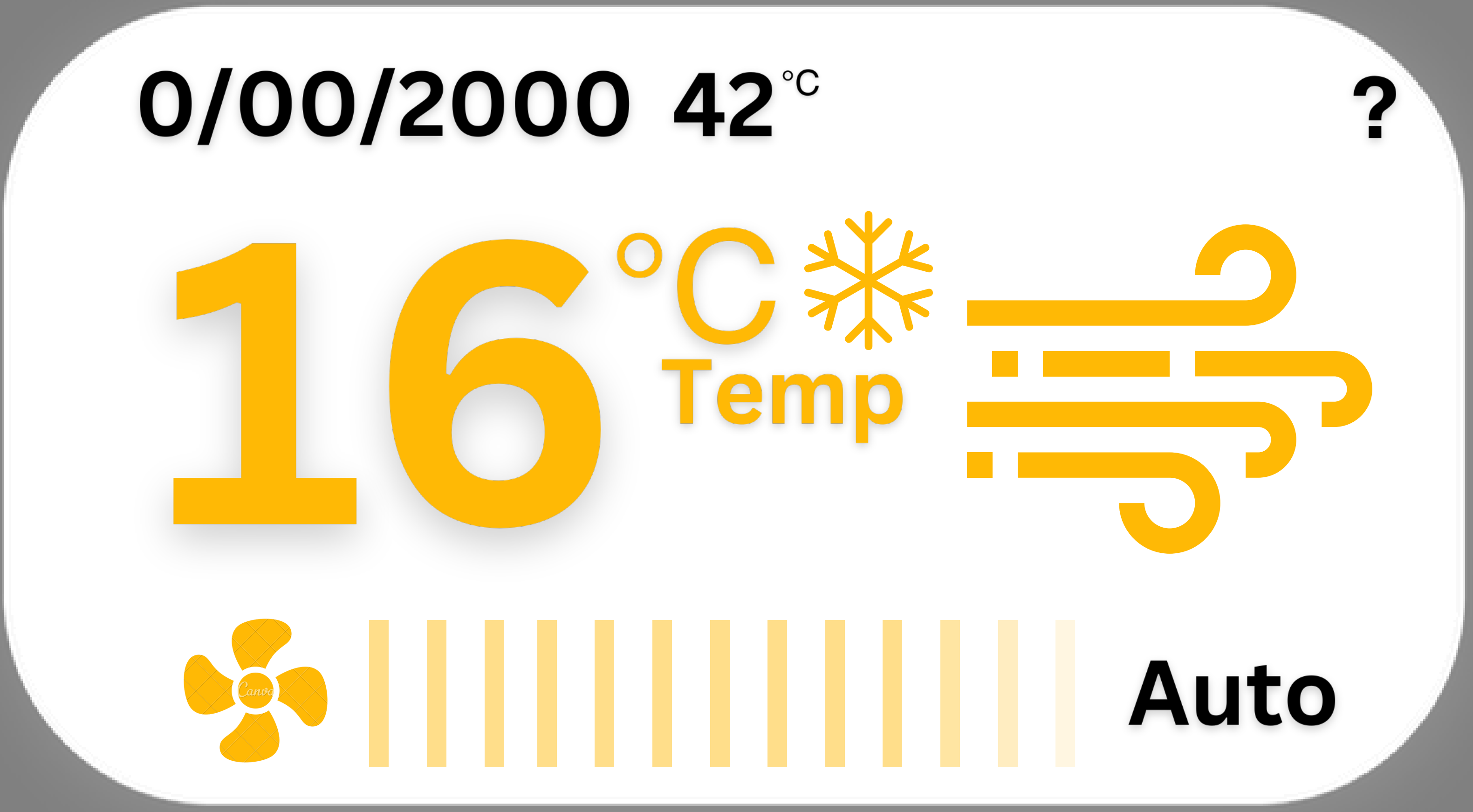
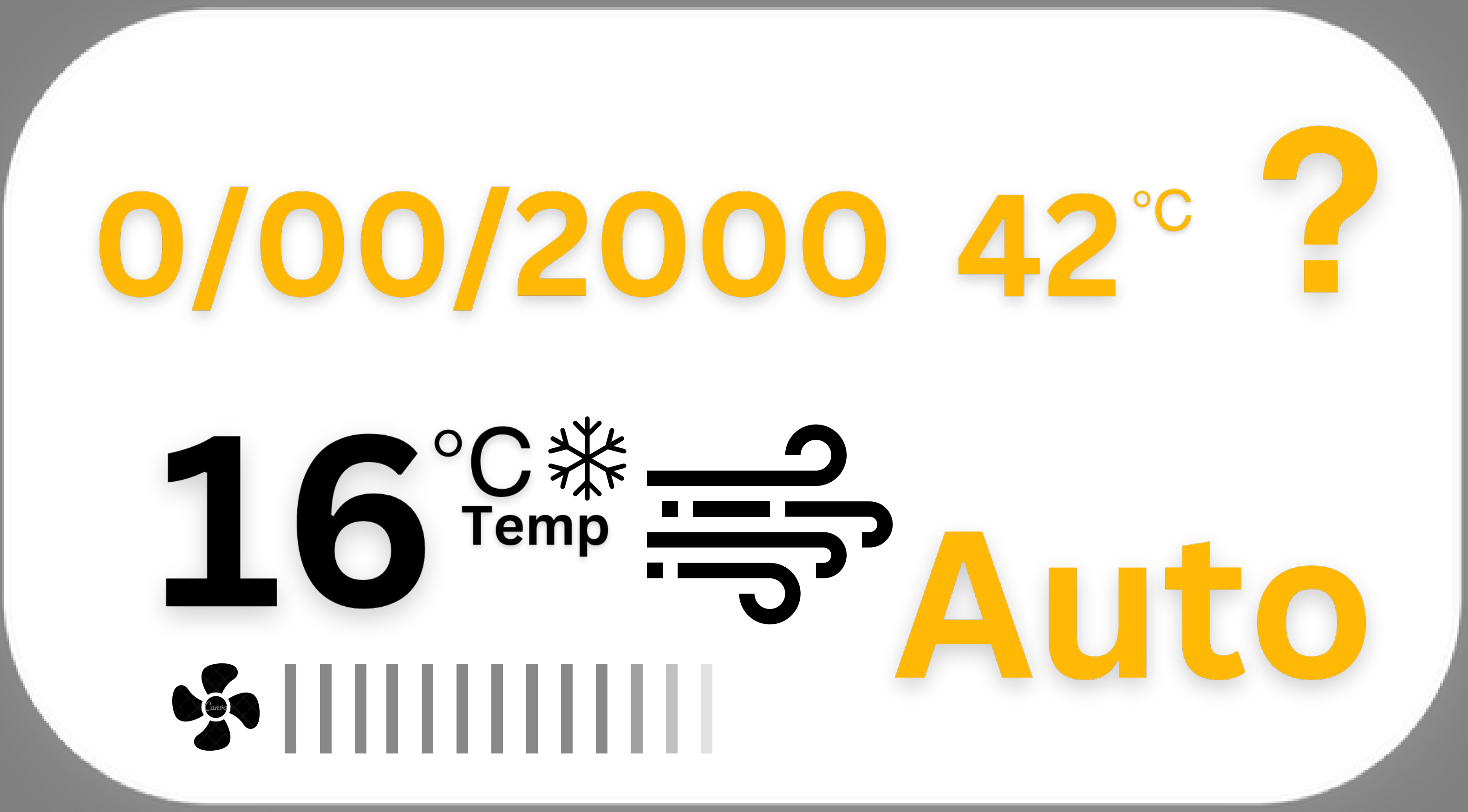


Temperature Display (16°C):

- Shows the current temperature setting of the AC.
- The snowflake icon indicates that the AC is in cooling mode.
- Wind symbol suggests fan speed or airflow direction.



- Date/Time (0/00/2000 42°C):
- Settings for real-time display
- Mode Indicator (Auto):
- The AC is set to automatically adjust settings like temperature and fan speed based on environmental conditions.
- ? for the user manual.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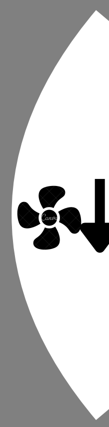


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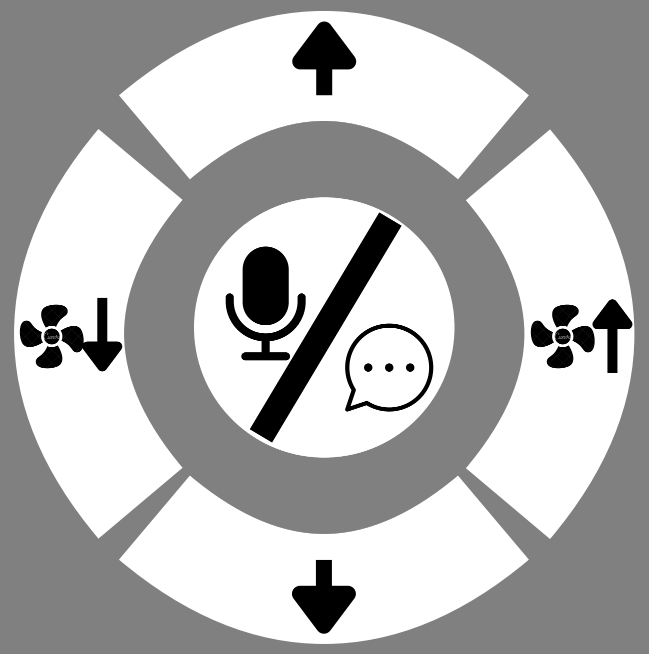


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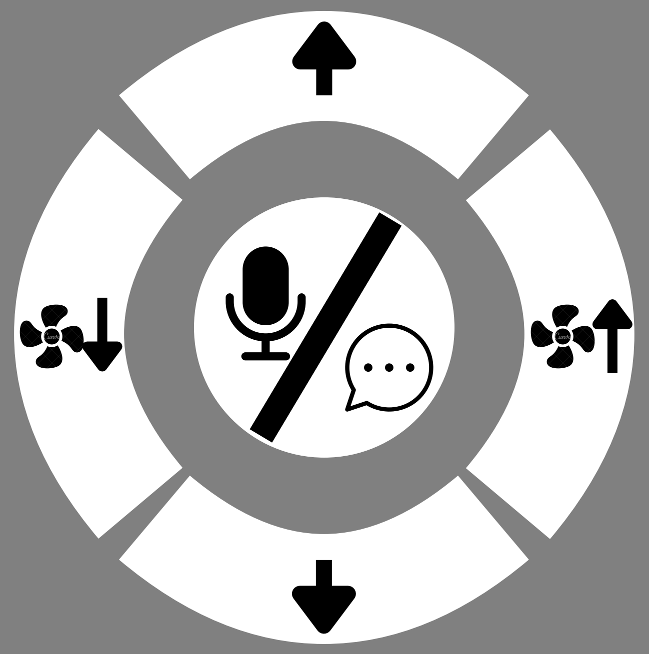

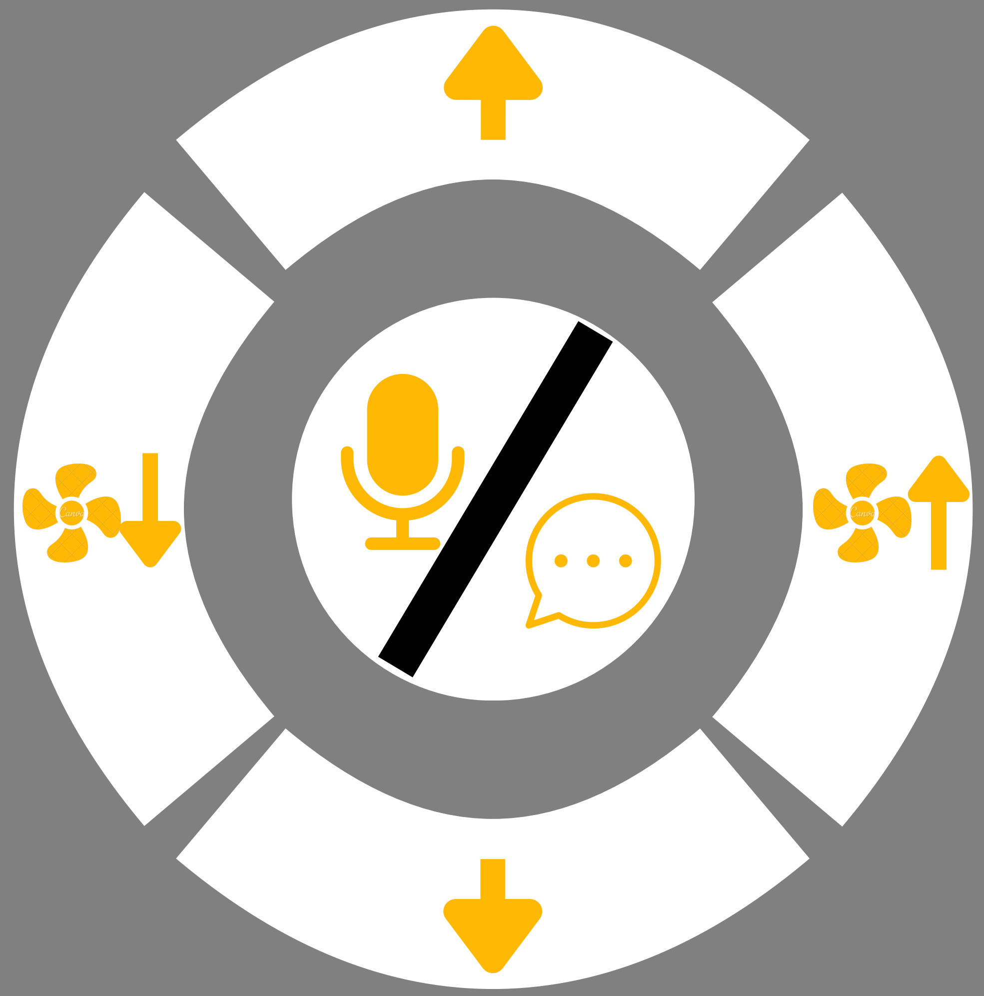

- Fan Control Wheel:
- Central circular control panel with directional arrows for adjusting airflow direction.
- Microphone symbol in the center suggests voice control functionality.
- Speech bubble indicates a messaging or feedback option.
- Fan Speed Indicator (Bars):
- Shows the current fan speed; the more bars, the higher the speed


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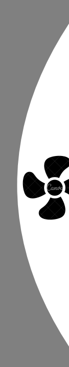

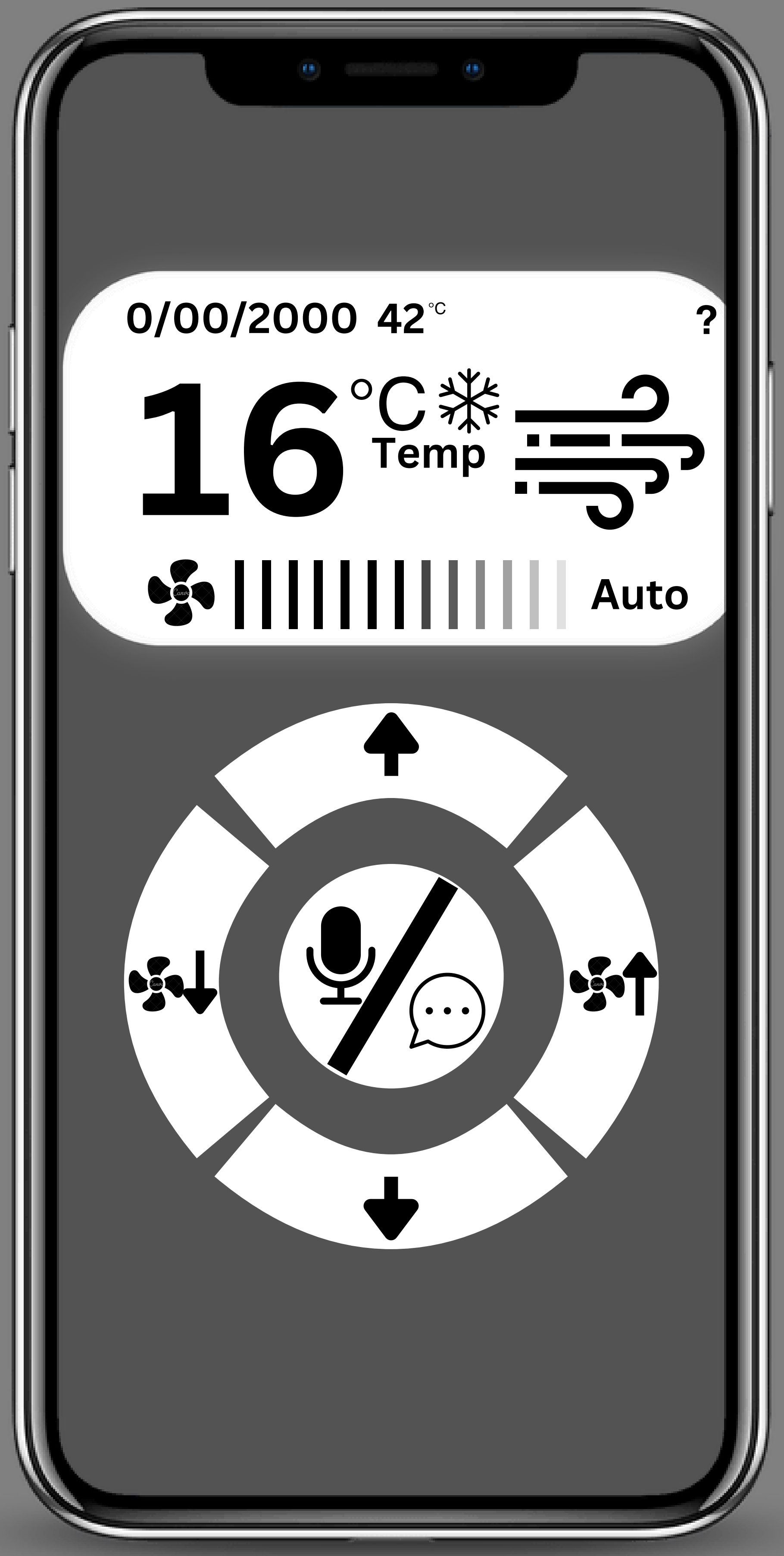
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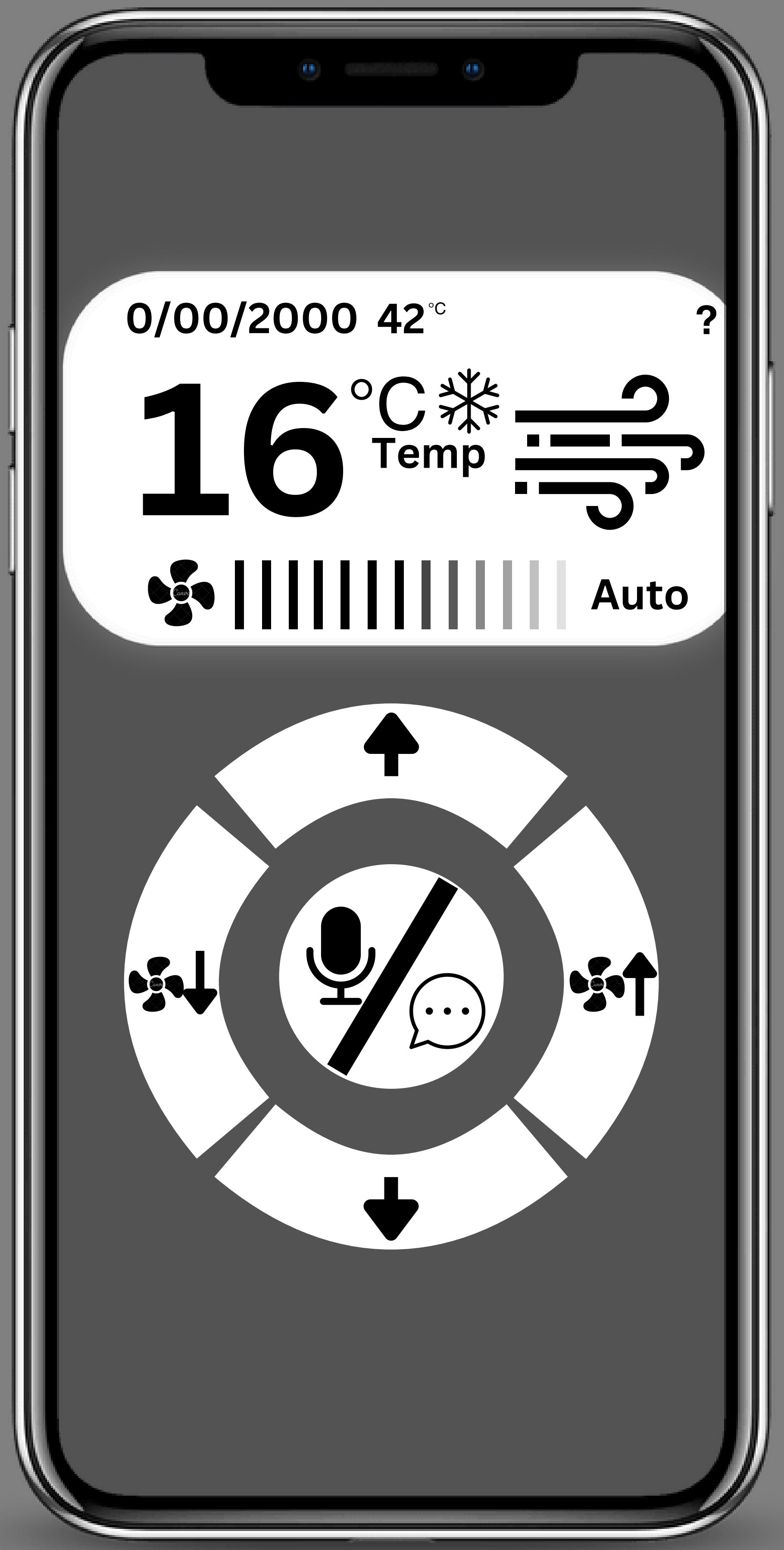
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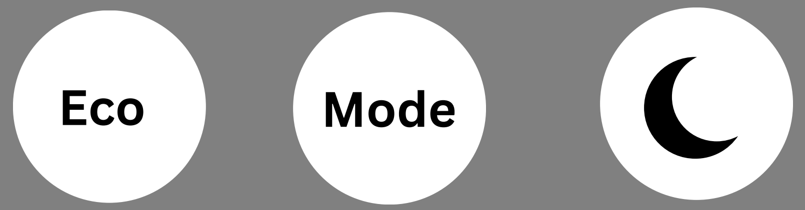
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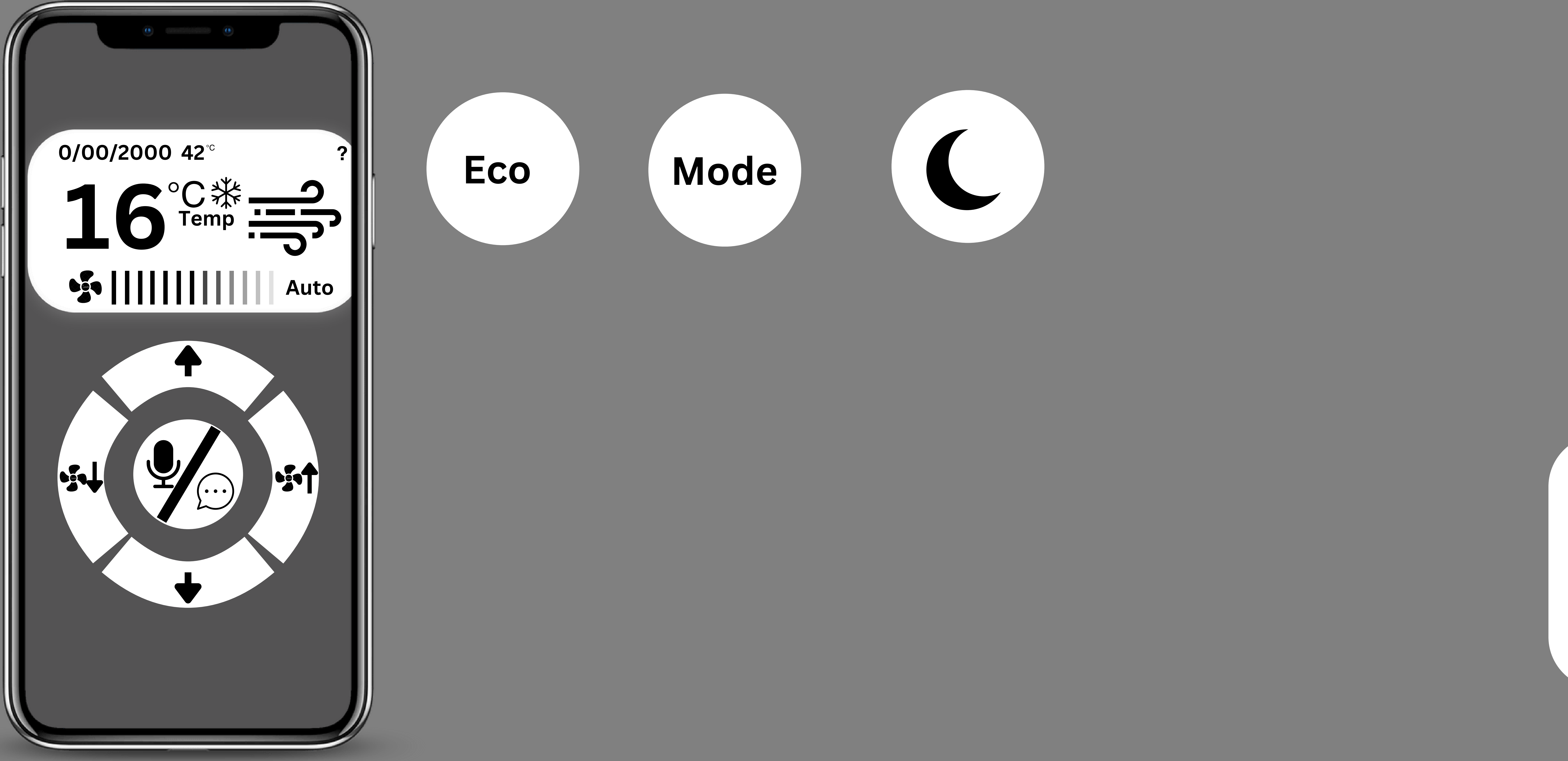
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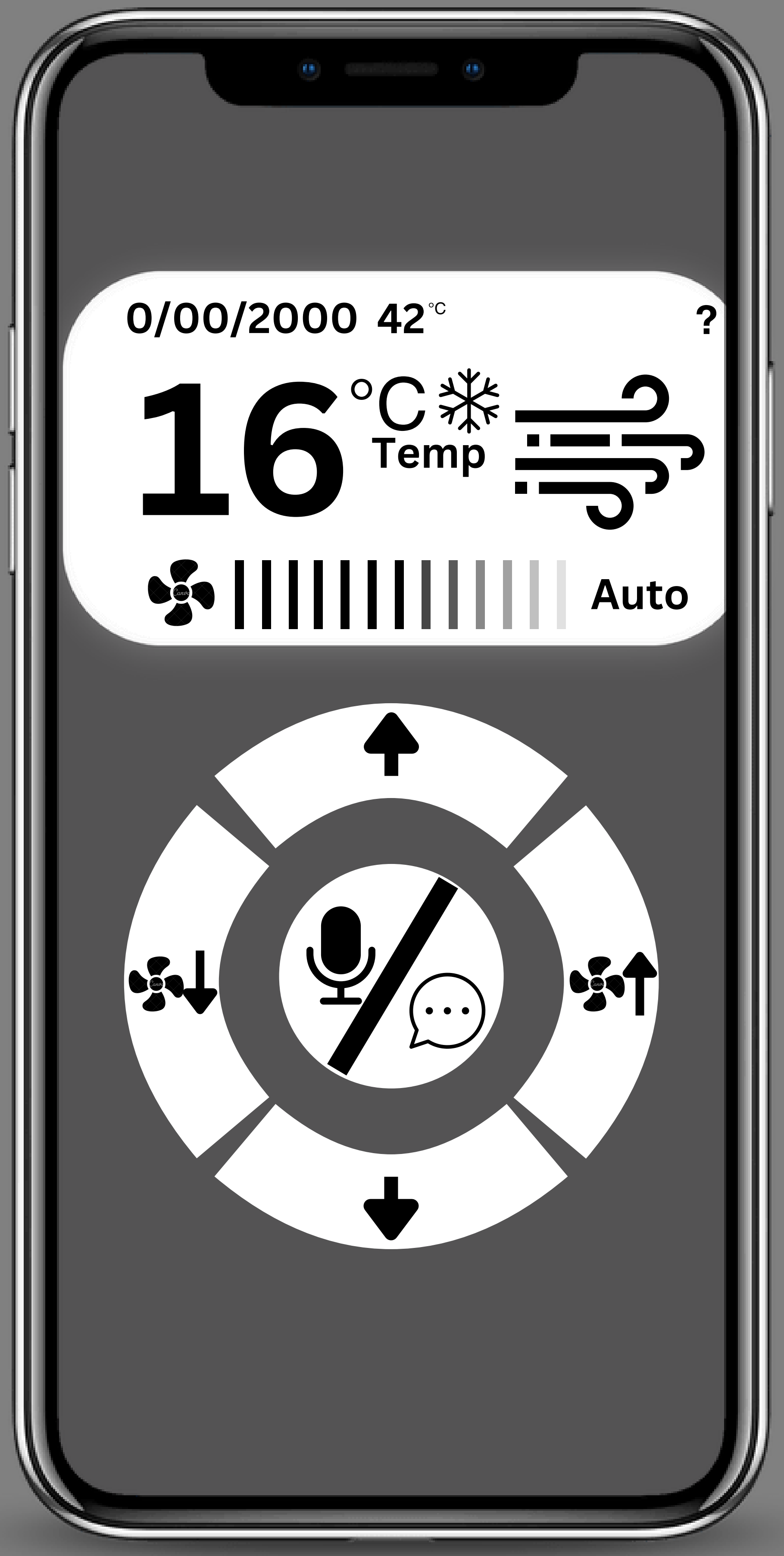
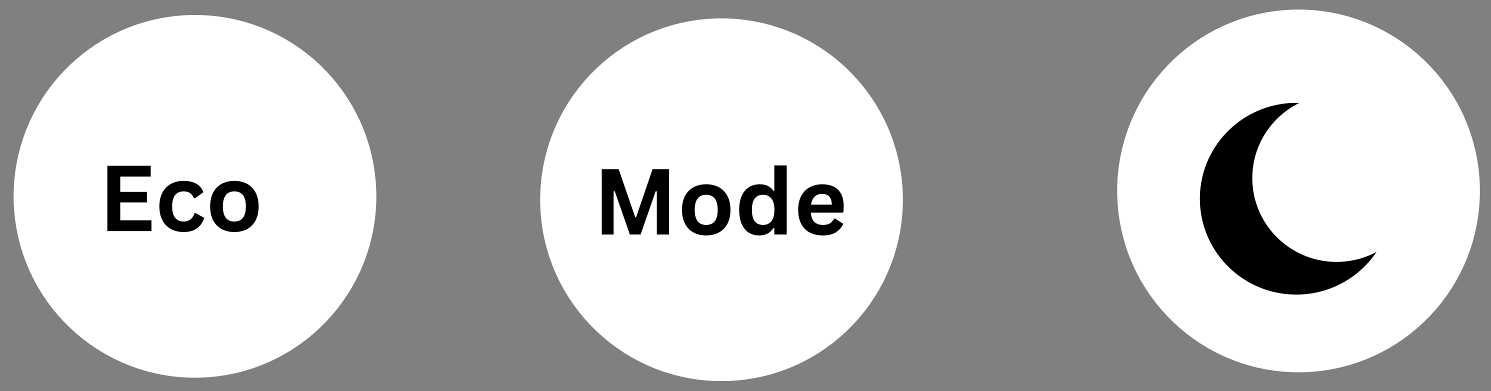
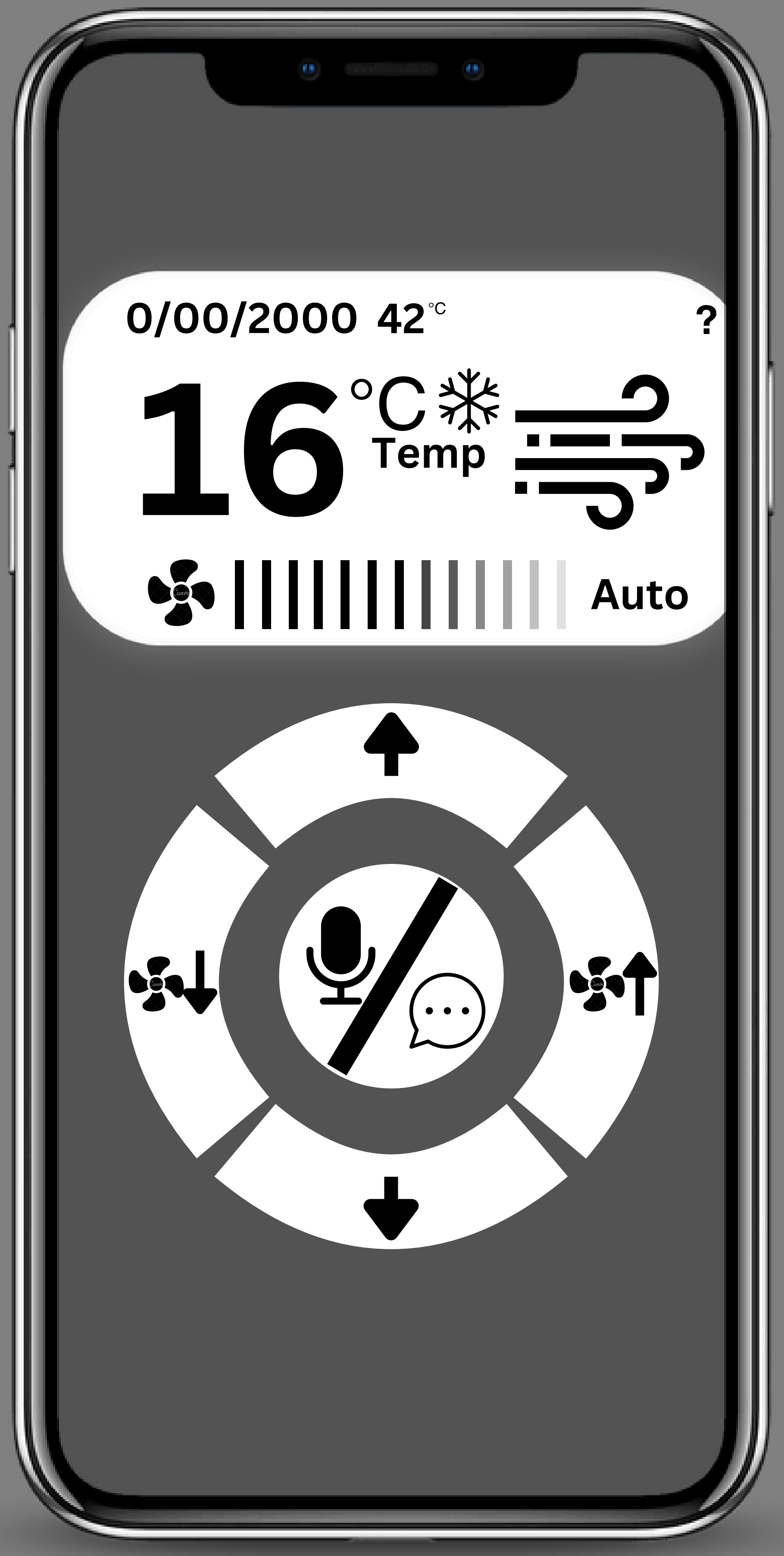




- Eco mode to enable the AC to operate using less energy.
- Mode is used to change between fan and cooling.
- Night mode to make the AC operate quietly.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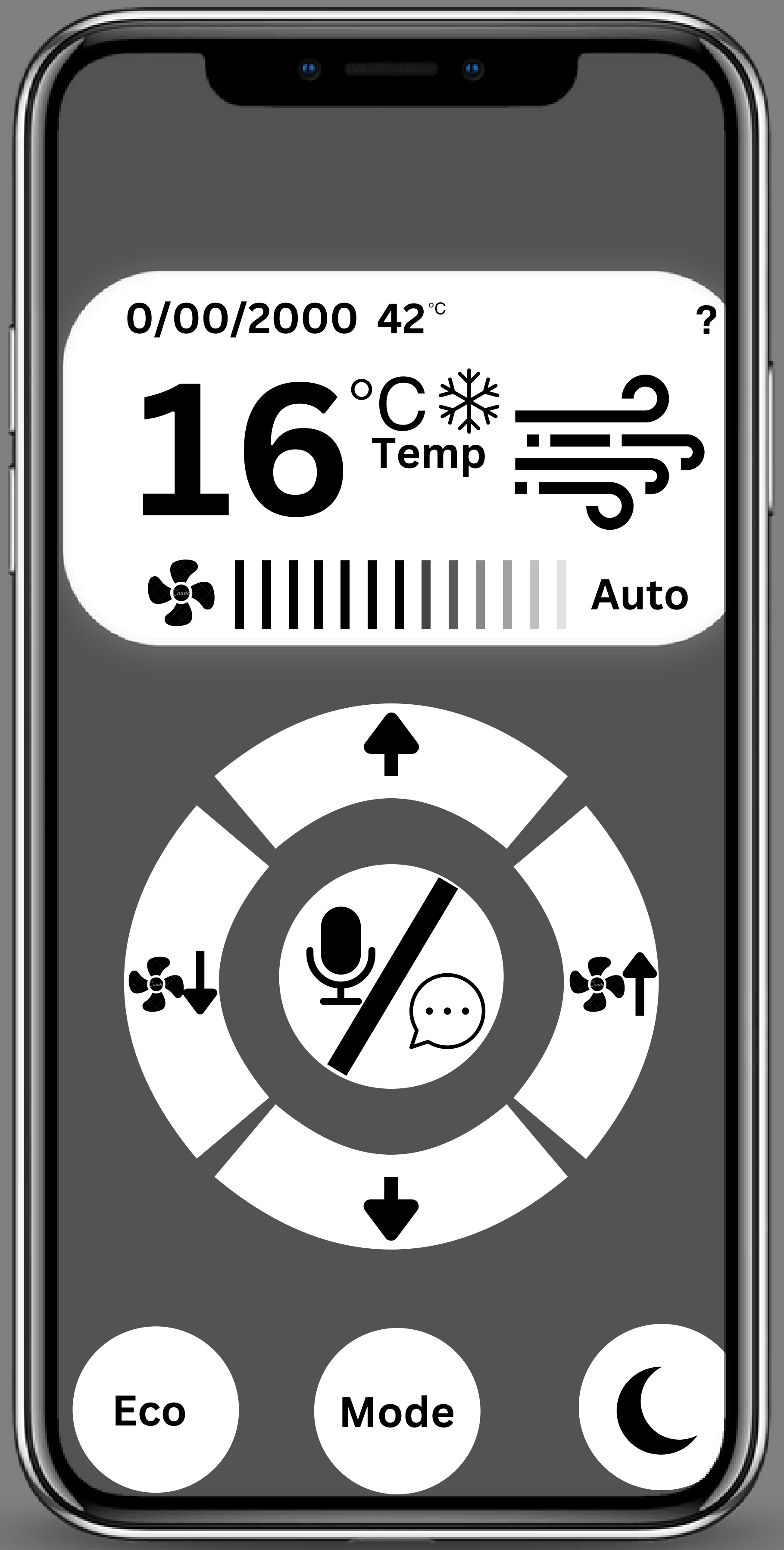
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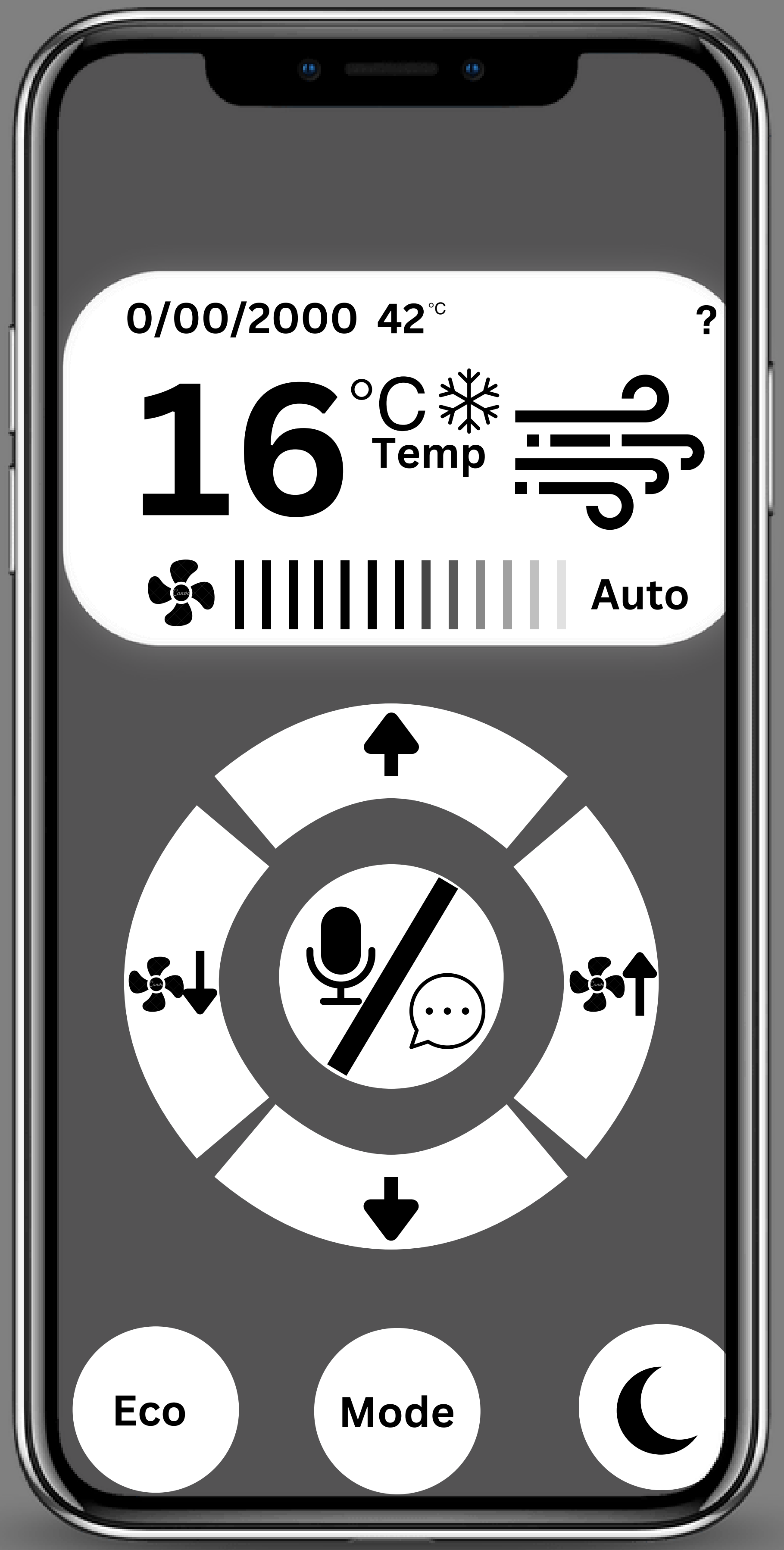
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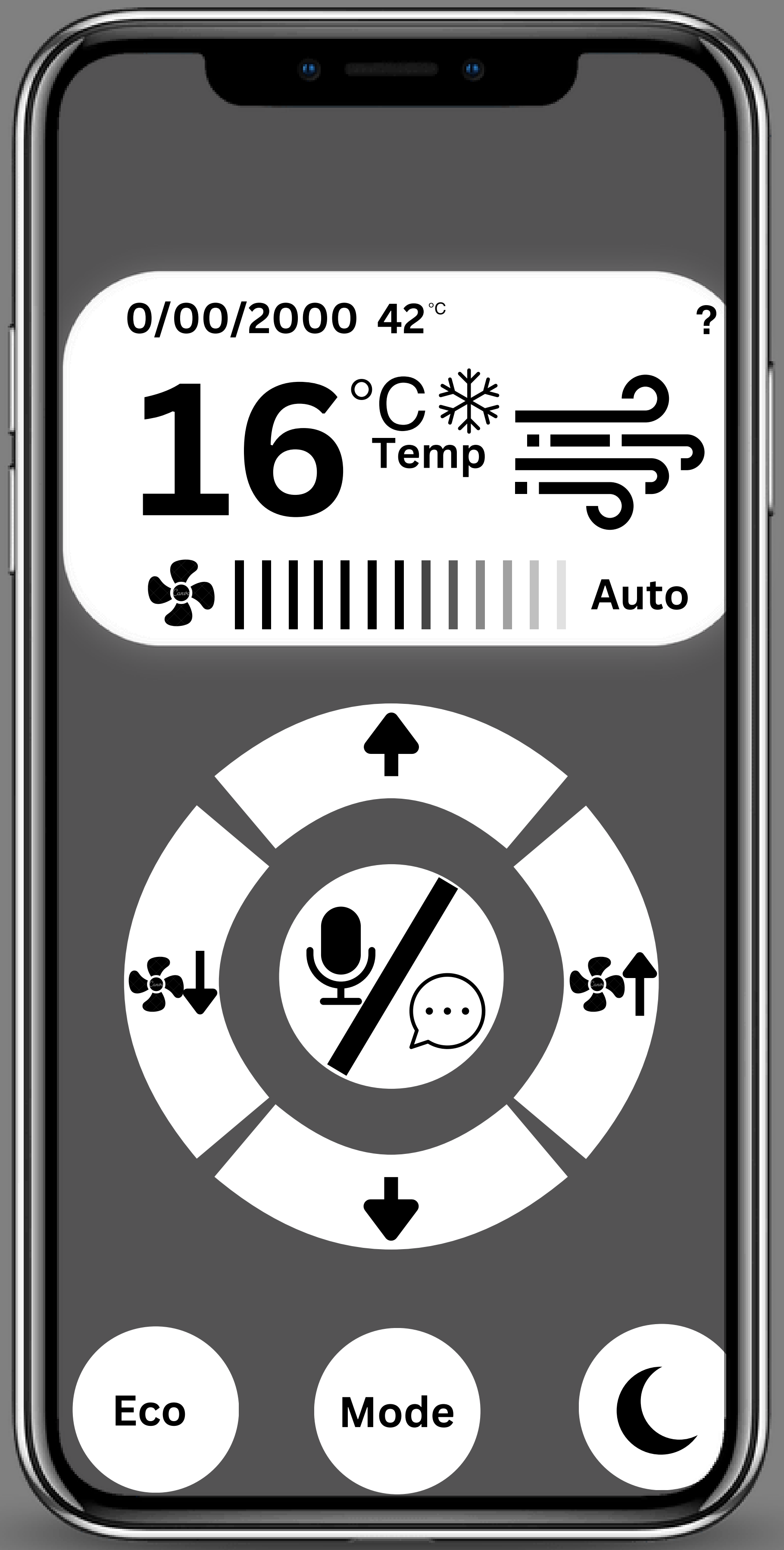
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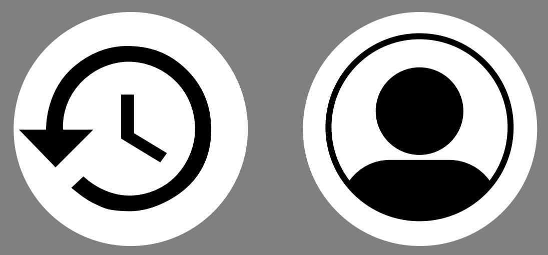
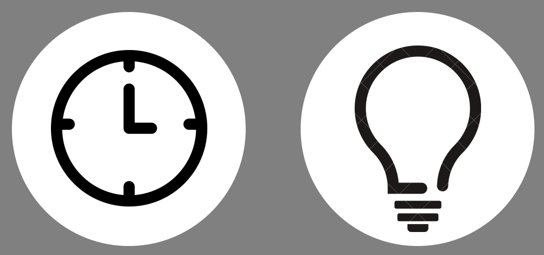
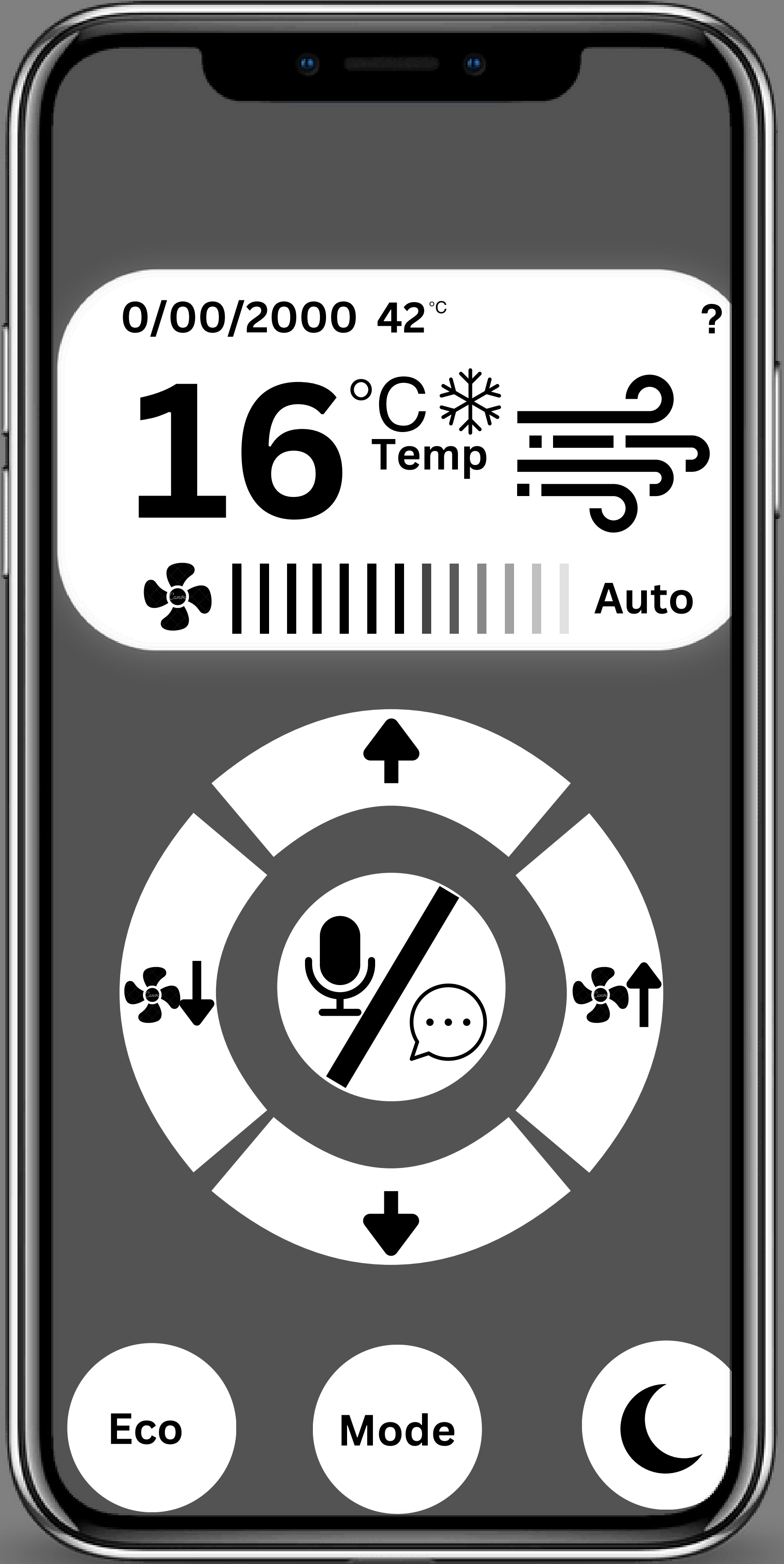



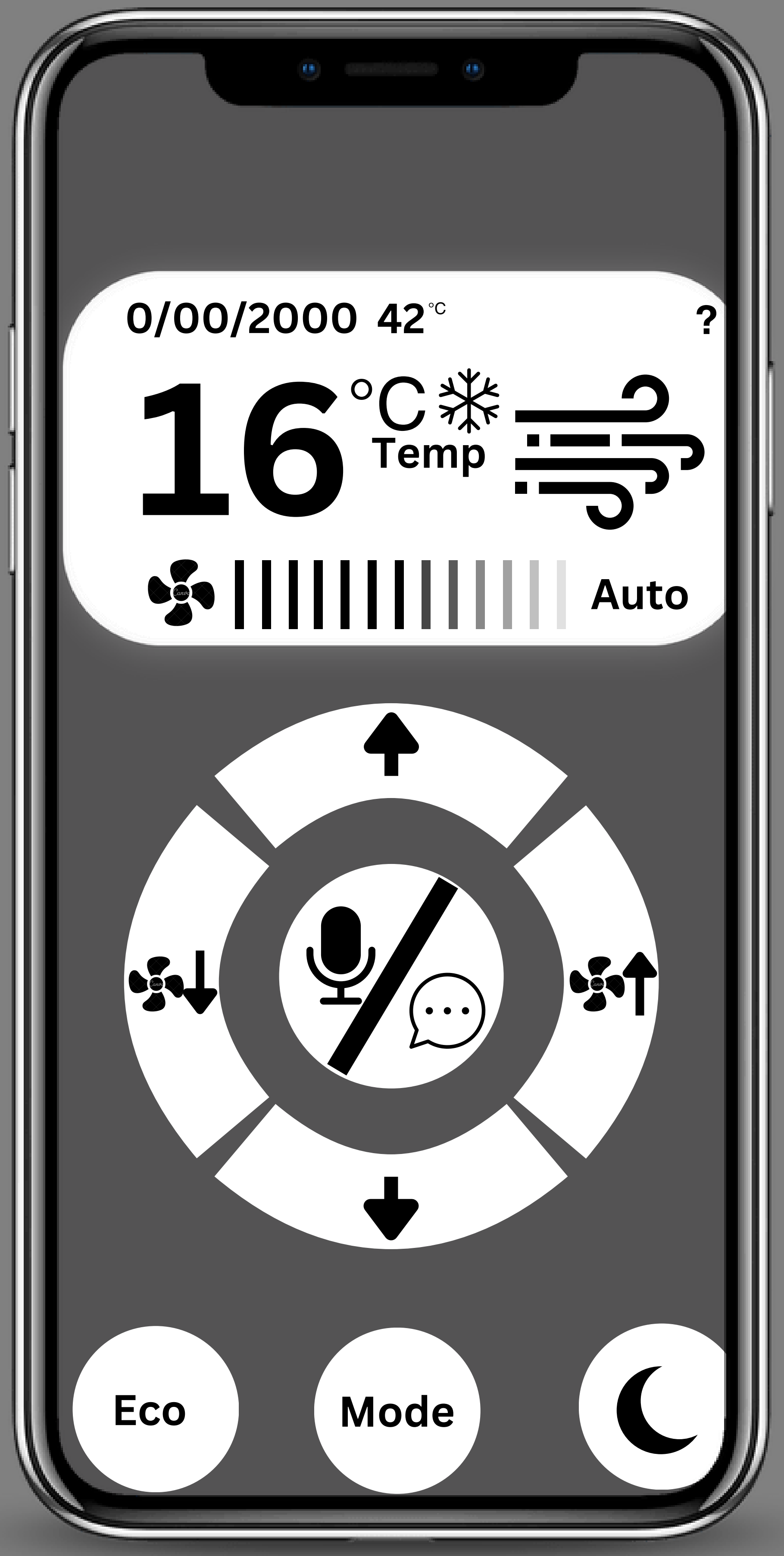


To track the history log of the AC.

The user’s account(used to connect other devices).

Lets the user be able to set a timer on a given date.

Lets the user be able to turn on lights.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CONCLUSION

A potential solution to the problem is to provide a system that allows employees to control lighting, temperature, and air quality. This system would include real-time sensors to monitor the environment and provide real-time updates. It would be a hybrid system designed to personalize their experience through user-friendly technology and interfaces, enhancing comfort and productivity. Additionally, it would foster inclusivity, reduce stress, and promote teamwork while improving energy efficiency in a workplace that prioritizes ethics and employee well-being.

REFERENCE

Duggal, N. (2024, August 13). What Are IoT Devices? Definition, Types, and 5 Most Popular for 2024. Simplilearn.com. https://www.simplilearn.com/iot-devices-article#:~:text=As%20of%202021%2C%20there%20were,54%20percent